

Project 3: Reinforcement Learning – Final Details

Dear RL Project Teams,

Here are the final details for Project 3. Please read the evaluation protocol and rules carefully.

General Guideline:

We do not impose restrictions on model architecture or training procedure. However, Task 2 and Task 3 must involve reinforcement learning components.

The policy must operate on visual observation of blocks, although the target locations are specified as (x, y, z) coordinates in the robot frame. You should make it easy to modify target locations during evaluation (in config file or as an argument, etc.). The wrist camera from SO101 robot is an RGB camera so if you want to use other image styles (gray scale, depth, etc.), you need to find ways to convert them. You are free to choose whichever representation yields better performance.

We encourage the use of goal-conditioned policies, where the policy receives task-specific inputs (e.g., target color or target location) and adapts its behavior accordingly. Both end-to-end and modular approaches (e.g., perception + control) are allowed, as long as the final behavior is governed by a learned policy. Training data may be obtained via simulation or teleoperation. Data collected from expert demonstrations in Task 1 may also be used as part of the RL training data, e.g., stored in a replay buffer for offline or hybrid RL methods.

Environment Setup:

For evaluation, the environment will be standardized:

- A light gray table (approximately #B8ADA9).
- Object with a fixed size and different but known colors.
- Bowls will be placed at randomized positions in the robot base frame (e.g., (x, y, z))
- Object positions will be randomized across rollouts

Students are expected to account for this variability during training.

Evaluation:

The project consists of three evaluation tasks, with a total of 150 points, plus an optional 50-point bonus.

Eval 1 (50 pts): Single Object Pick-and-Place

A single wooden block is placed randomly on the table. The policy must pick up the block and place it into a bowl.

- The color of the block is not constrained
- The target bowl position is specified
- 5 rollouts will be performed

Each successful rollout yields 10 points. Success is defined as: the block is placed inside the bowl and released

Both BC and RL-based methods are allowed for this task.

Eval 2 (50pts): Targeted Pick-and-Place in Clutter

Two blocks of different colors are placed adjacent to each other (flat cluster). One target color will be specified.

The policy must identify the block of the specified color, grasp it, and place it into a specified bowl.

- The target color is provided as input
- The target bowl position is specified
- 5 rollouts will be performed

Each successful rollout yields 10 points. Success requires that the correct block is placed inside the bowl and released.

This task must involve reinforcement learning. Expert data collected from teleoperation is encouraged to use for training efficiency.

Eval 3 (50pts): Sequential Multi-Step Pick-and-Place

Four blocks of distinct colors are placed in the workspace.

The policy is given a sequence of three pick-and-place goals, each specifying:

- A target color
- A target bowl position

The bowl positions remain fixed within each rollout.

The policy must execute the sequence in order.

Additionally:

- Policy switching and/or switching perception modules (e.g., color filters) is allowed
- Interaction with non-target blocks (e.g., pushing or rearranging) is permitted and encouraged if it facilitates task completion
- 5 rollouts will be performed

Scoring per rollout:

- 4, 4, and 2 pts for completing each step
- Partial credit is awarded for partial completion

A step is considered successful if the correct block is placed inside the corresponding bowl and released.

This task must involve reinforcement learning. Expert data collected from teleoperation is encouraged to use for training efficiency.

Bonus (50pts): Efficiency or Singulation

The bonus will be awarded based on one of the following criteria:

Option A: Speed

Among rollouts in Eval 3 that successfully complete all three steps, a rollout is considered successful if it is completed within a specified time limit. Bonus points will be awarded based on the number of successful rollouts.

Option B: Singulation

Given a stack or clustered arrangement of three or four blocks, the policy must separate them into individually graspable configurations.

Further details on bonus evaluation will be clarified before final evaluation.

Additional Notes:

- No additional hardware, objects, or modifications to the setup are allowed

References and Resources:

- SO101 Tutorial: [SO-101](#)
- SO101 in Isaac Lab: https://github.com/MuammerBay/isaac_so_arm101
- SO101 in MuJoCo: <https://github.com/RobotControlStack/robot-control-stack>

Best Regards

Project 3 TAs